Project: Worked examples
Title: Fillet weld (EN 1993-1-8 simplified method)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A fillet weld carrying forces parallel to, transverse to and normal to its axis, by the simplified method: the resultant over the throat area against the design shear strength of the weld, with the required throat and leg.

WHAT IT CALCULATES	RELATION
Resultant force on the weld	$F = \sqrt{(F_{\parallel}^2 + F_{\perp}^2 + F_n^2)}$
Design shear strength of the weld, simplified method	$f_{vw,d} = f_u/(\sqrt{3}\cdot\beta_w\cdot\gamma_{M2})$
Shear stress on the throat	$\tau_w = F/(a\cdot L)$
Weld resistance and utilisation	$F_{w,Rd} = f_{vw,d}\cdot a\cdot L$; $u = F/F_{w,Rd}$
Throat needed (not less than 3 mm) and the matching leg	$a_{req} = \max(F/(f_{vw,d}\cdot L), 3\ \text{mm})$; $\text{leg} = \sqrt{2}\cdot a$

Basis: Simplified method for design resistance of fillet weld[1](4.5.3.3); Effective throat thickness: minimum 3 mm[1](4.5.2(2)); Effective length: welds shorter than 30 mm or 6 throats not to carry load[1](4.5.1(2)); Long joints[1](4.11).

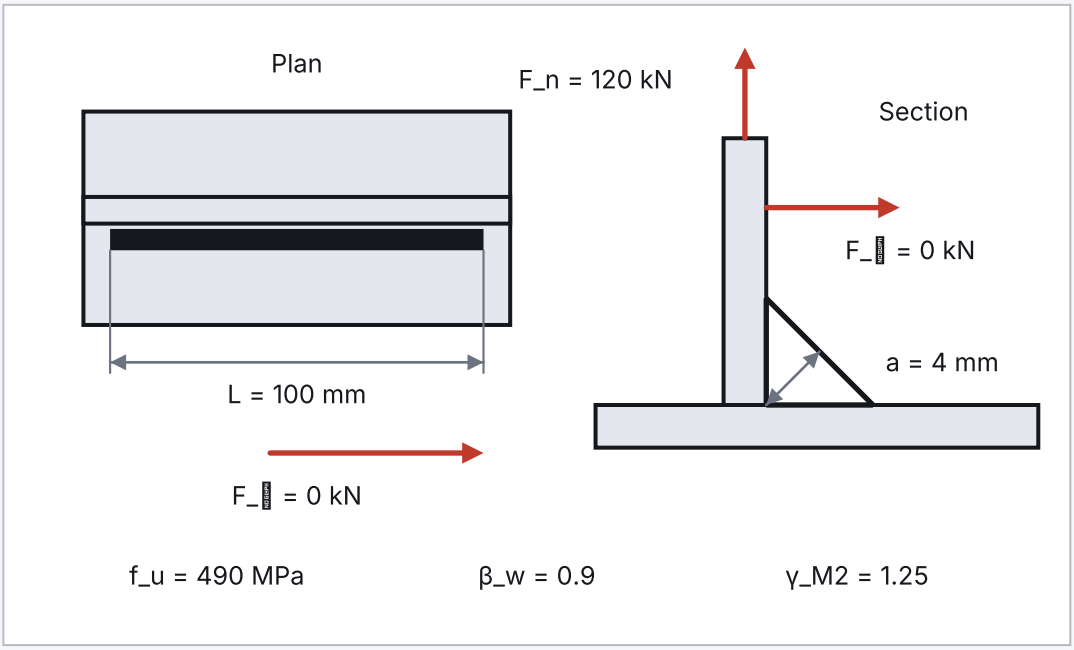


Figure 1 — A fillet weld of effective length L: in plan, the force along it; in section, the throat a and the forces across it and normal to the plate. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_par	Force parallel to the weld axis	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
F_tr	Force transverse to the weld axis, in the plane of the parts	0	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_n	Force normal to the plane of the parts	120	kN	spec: LOAD-SCHEDULE-01	-100000 … 100000 kN
L	Effective weld length	100	mm	drawing: DRG-WD-702 rev A	30 … 10000 mm
a	Throat thickness	4	mm	drawing: DRG-WD-702 rev A	3 … 50 mm
f_u	Ultimate strength of the weaker part joined	490	MPa	spec: SPEC-STR-02 design criteria	300 … 1000 MPa
beta_w	Correlation factor β_w (0.8 S235, 0.85 S275, 0.9 S355, 1.0 S420/S460)	0.9		spec: SPEC-STR-02 design criteria	0.8 … 1
gamma_M2	Partial factor γ_M2	1.25		spec: SPEC-STR-02 design criteria	1 … 2

3. Assumptions

A1. The resultant force is shared uniformly along the effective length. (Simplified method.)
A2. Equal-leg fillet: leg = √2 × throat. (Geometry.)

4. Working

```
[9] F = sqrt(F_par^2 + F_tr^2 + F_n^2)
= sqrt((0 kN)^2 + (0 kN)^2 + (120 kN)^2)
= 120 kN
```

Resultant force on the weld

```
[10] f_vw_d = f_u / (sqrt(3) * beta_w * gamma_M2)
= 490 MPa / (sqrt(3) × 0.9 × 1.25)
= 251.47 MPa
```

[1](4.5.3.3)

Design shear strength of the weld

```
[11] tau_w = F / (a * L)
= 120 kN / (4 mm × 100 mm)
= 300 MPa
```

Stress on the throat

```
[12] F_w_Rd = f_vw_d * a * L
= 251.47 MPa × 4 mm × 100 mm
= 100.59 kN
```

[1](4.5.3.3)

Design resistance of the weld

```
[13] u = F / F_w_Rd
= 120 kN / 100.59 kN
= 1.193
```

Utilisation

```
[14] a_calc = F / (f_vw_d * L)
= 120 kN / (251.47 MPa × 100 mm)
= 4.772 mm
```

Throat needed for the load

```
[15] a_req = max(a_calc, 3 mm)
= max(4.772 mm, 3 mm)
= 4.772 mm
```

[1](4.5.2(2))

Required throat (not less than 3 mm)

```
[16] s_leg = sqrt(2) * a
= sqrt(2) × 4 mm
= 5.6569 mm
```

Leg length of the throat provided (equal legs)

5. Results

SYMBOL	VALUE	DESCRIPTION
F	120 kN	Resultant force
f_vw_d	251.47 MPa	Design shear strength
tau_w	300 MPa	Throat stress
F_w_Rd	100.59 kN	Weld resistance
u	1.193	Utilisation
a_calc	4.772 mm	Throat needed
a_req	4.772 mm	Required throat
s_leg	5.6569 mm	Leg length provided

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Weld resistance	u = 1.193	u ≤ 1	1.19	[1](4.5.3.3)

7. Applicability

No warnings.

8. Verification

Template struct.fillet-weld@1.0.1 Content hash fc8699cc96f9dfc1 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.